Quantum Mechanics II — Exercise Sheet 0

Operators and their domains

Variant: Questions only

Exercise 1 (A domain changes the adjoint). On $\ell^2(\mathbb{N})$ let A act by $(Ac)_n = n^2 c_n$ on $D(A) = \{c : \sum_{n\geq 1} n^4 |c_n|^2 < \infty\}$. Let B have the same action on $D(B) = \{c \in D(A) : c_1 = 0\}$. (i) Decide which operator is densely defined. For the densely defined operator, determine its adjoint and decide whether it is self-adjoint. (ii) Explain why the lecture's definition of an adjoint does not apply to the other operator.

Exercise 2 (A bounded function of position). On $L^2(\mathbb{R})$ consider multiplication by $g(x) = \arctan x$, on its maximal domain. (i) Determine the domain and the operator norm. (ii) Decide whether the operator is self-adjoint and whether it has eigenvalues. Use only Lecture 0, without a claim about its spectrum.

Exercise 3 (Two multiplication domains). Let Q be multiplication by x on its maximal domain in $L^2(\mathbb{R})$. For $a > 1/4$ let $f_a(x) = (1 + x^2)^{-a}$. Determine exactly when $f_a \in D(Q)$ and when $f_a \in D(Q^2)$. Include the threshold cases and give a vector in $D(Q) \setminus D(Q^2)$.

Exercise 4 (Unitary equivalence includes the domain). On $L^2(\mathbb{R})$ let $(Jf)(x) = f(-x)$ and let M_+ be multiplication by e^x on its maximal domain. Find $J^{-1} M_+ J$, including its domain. Check whether $f(x) = e^{-|x|}$ belongs to that domain.